

Syntrix – Personal Digital Vault

2. Full Technical Stack Document

Project: Personal Digital Vault
Repository: viratmission/Personal-Digital-Vault
Default branch reviewed: main

1. Technology Stack Summary

Category	Technology used
Backend	ASP.NET Core Web API on .NET 8 (net8.0)
Frontend	Angular 19.2.x
Database	Microsoft SQL Server / LocalDB (PersonalDigitalVaultDB)
Languages	C#, TypeScript, HTML, CSS
ORM / Data Access	Entity Framework Core 8 – SQL Server provider
Authentication	JWT Bearer authentication
Password Security	ASP.NET Core Identity PasswordHasher<T>
API Documentation	Swagger / OpenAPI via Swashbuckle.AspNetCore 6.6.2
Encryption	AES-256 for protected credential/document data
File Security	File hashing and file encryption services
UI Styling	Tailwind CSS 4.x
Client HTTP	Angular HttpClient with authentication interceptor
Reactive / Runtime	RxJS 7.8, Zone.js 0.15
Testing tooling	Jasmine / Karma (Angular project dependencies)
Development tools	Visual Studio / VS Code, .NET SDK, Node.js/npm, Git/GitHub

2. .NET Version

The backend project targets **.NET 8** using the TargetFramework value **net8.0**. The project uses ASP.NET Core Web SDK.

3. Backend Technology / Framework

The backend is an ASP.NET Core Web API. Controllers expose REST endpoints. Dependency Injection registers repositories and services including authentication, folders, documents, credentials, search, administration, AES encryption, file encryption and hashing.

4. Frontend Technology / Framework

The frontend is an Angular 19 application. The package configuration specifies Angular 19.2.x, Angular Router, Angular Forms and Angular HttpClient. Tailwind CSS 4.x is included for UI styling. An HTTP interceptor is configured to attach authentication-related requests.

5. Database

The application uses SQL Server through Entity Framework Core 8. The configured development connection uses SQL Server LocalDB with database name **PersonalDigitalVaultDB**. The DbContext contains Users, Folders, Documents and Credentials sets, with EF Core migrations present in the repository.

6. APIs / Services

Key application services include authentication/register/login, profile handling, folder management, document management, credential management, search, admin functions, AES encryption, file encryption and file hashing. Swagger/OpenAPI is configured for API exploration. JWT Bearer is used for protected API authorization.

7. Security Technologies

JWT tokens are signed with HMAC-SHA256 and validated for issuer, audience, lifetime and signing key. Passwords are stored as hashes using PasswordHasher<User>. AES-256 is implemented for encrypted values; the AES key is expected from configuration/user secrets rather than hard-coded in the encryption service.

8. Important Libraries / Packages

Microsoft.AspNetCore.Authentication.JwtBearer 8.0.0; Microsoft.EntityFrameworkCore.SqlServer 8.0.0; Microsoft.EntityFrameworkCore.Tools 8.0.0; Swashbuckle.AspNetCore 6.6.2; Angular 19.2.x; Tailwind CSS 4.3.x; RxJS 7.8.x; TypeScript 5.7.x.

Source verification

This document was prepared from the project's GitHub source files, including PersonalDigitalVault.csproj, Program.cs, AppDbContext.cs, AesEncryptionService.cs and frontend/package.json.